Trader Behavior vs Market Sentiment Summary

Methodology

The analysis integrates Bitcoin Fear/Greed sentiment data with Hyperliquid trader execution records at a daily level. After standardizing timestamps and aligning datasets, daily trader-level features were created, including PnL, trade frequency, exposure (USD size), and directional bias. Traders were segmented by exposure and clustered based on profitability and volatility to identify behavioral archetypes

---

Key Insights

1. **Exposure Dominates Profitability**
   High-exposure traders generate 10–30x higher average PnL than low-exposure traders across all sentiment regimes.
2. **Greed Regimes Amplify Returns**
   The highest profitability occurs during Greed periods, particularly for aggressive traders.
3. **Distinct Behavioral Archetypes Exist**
   Clustering reveals:

1. Ultra-aggressive traders with extreme volatility
2. Disciplined aggressive traders with better risk-adjusted returns
3. Low-intensity retail traders with modest performance

Directional Bias Shifts with Sentiment
Traders exhibit stronger short bias during Greed regimes, suggesting contrarian positioning behavior.

Strategy Recommendations

1. **Sentiment-Conditioned Exposure Scaling**
   Increase allocation to disciplined aggressive traders during Greed regimes, where return amplification is strongest.
2. **Volatility Participation Filter**
   During Fear regimes, restrict capital allocation to high-volatility traders with proven profitability while limiting exposure for low-impact participants.

Bonus:A simple predictive model using sentiment and behavioral features shows modest predictive power for next-day profitability, suggesting limited but present signal from sentiment-conditioned behavior.